



**INSTITUT  
POLYTECHNIQUE  
DE PARIS**

# Assignment

**Linear Time Series**  
Master-I Applied Mathematics and Statistics

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# 1 The Data

## 1.1 Series Description

We selected a French Industrial Production Index (IPI) series from INSEE's database, focusing on the sub-sector "Preparation of fruit and vegetable juices" (NAF Rev. 2, Class 10.32). The series is seasonally and working-day adjusted (CVS-CJO), with monthly frequency and indexed to base year 2021 = 100.

The chosen series reflects the production activity within the French industrial subsector responsible for processing fruits and vegetables into juice products. This index provides insight into the sector's output dynamics over time. As it exhibits multiplicative growth patterns and increasing variance, a logarithmic transformation was considered appropriate to stabilize variance and interpret changes in percentage terms.

We started off by firstly cleaning the raw data by removing metadata rows, converting dates to yearmon format, filtering to retain only validated observations (Code 'A'), and ensuring numeric values. This ensured consistency for time series conversion. And then we used this cleaned data to transform it into a time series.

## 1.2 Transforming the Series to Achieve Stationarity

Before proceeding with ARMA modeling, we must ensure that the series is stationary. Non-stationary time series often exhibit trends or changing variance over time, which violates the assumptions of many time series models.

### 1.2.1 Visual Inspection of Original Series

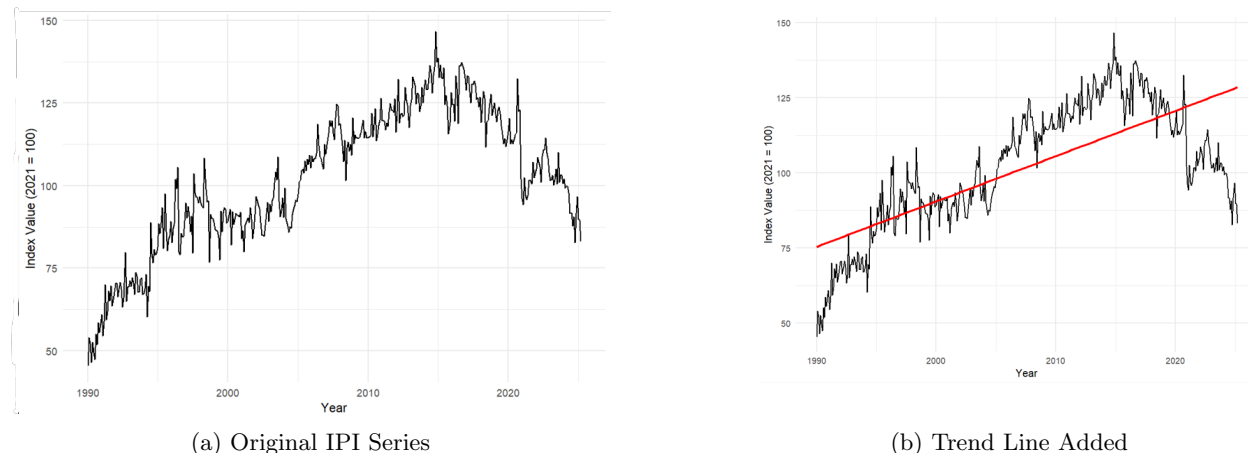


Figure 1: Time plots of the original IPI series showing upward trend and increasing variance.

From the plots above, it is evident that the original series has an upward trend and increasing variability over time — both indicators of non-stationarity.

We also plotted the seasonal pattern trend which revealed recurring patterns across months, suggesting potential seasonality in the data.

## 1.3 Stationarity

### 1.3.1 Stationarity Check

We performed the Augmented Dickey-Fuller (ADF) test to statistically assess stationarity:

ADF Test Results:

Original series p-value: 0.9763

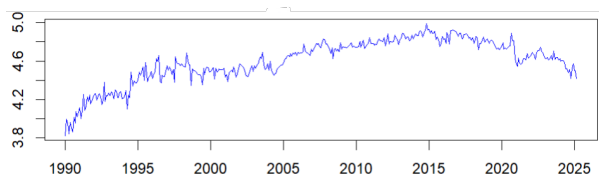
Differenced series p-value: 0.0132

Since the p-value for the original series is greater than 0.05, we fail to reject the null hypothesis — the series is non-stationary. After applying a log transformation and first differencing, the transformed series yields a p-value less than 0.05, indicating stationarity.

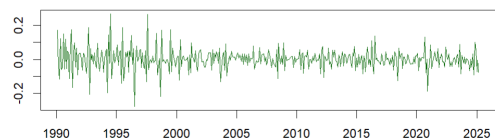
## 1.4 Transformation Steps

To achieve stationarity, we applied the following transformations:

- **Log Transformation:** Stabilized the variance of the series.
- **First Differencing:** Removed the stochastic trend.



(a) Log-transformed series

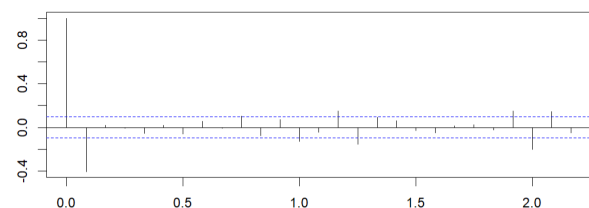


(b) First difference of log-transformed series

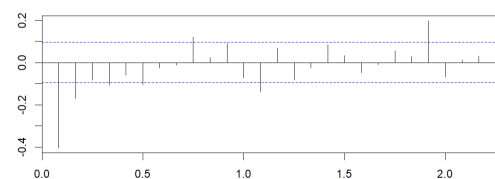
Figure 2: Transformed series showing stationarity.

We applied a log transformation to stabilize variance (evident from increasing variability in Figure 1). While first differencing addressed the stochastic trend (ADF p-value improved from 0.98 to 0.01).” These transformations successfully produced a stationary version of the series suitable for further modeling.

### 1.4.1 ACF and PACF of Transformed Series



(a) ACF of differenced log series



(b) PACF of differenced log series

Figure 3: Autocorrelation and partial autocorrelation functions of the stationary series

These plots will be used in Part II to identify the order of the ARMA model.

## 2 ARMA Models

### 2.1 Model Identification

We began by analyzing the ACF and PACF of the stationary differenced log series ( $\nabla \log(X_t)$ ). The plots (Figure 3) suggested:

- **ACF** decays gradually  $\Rightarrow$  MA(1) component
- **PACF** cuts off after lag 1  $\Rightarrow$  AR(1) component

This pattern suggests that the series may follow an AR(1) process. Similarly, the ACF showed a cut-off at lag 1, indicating a possible MA(1) component.

Based on these observations, we considered three candidate models: ARMA(1,1), ARMA(2,1) and ARMA(1,2)

### 2.2 Model Estimation

We estimated each candidate ARMA model using the `Arma()` function from the `forecast` package: ARIMA(1,0,1), ARIMA(2,0,1) and ARIMA(1,0,2)

The models were then compared based on AIC and BIC:

AIC and BIC Comparison:

|            | df | AIC       | BIC     |
|------------|----|-----------|---------|
| fit_arma11 | 4  | -1253.380 | -136.54 |
| fit_arma21 | 5  | -1252.969 | -132.15 |
| fit_arma12 | 5  | -1252.212 | -129.92 |

Among all tested models, **ARIMA(1,0,1)** had the lowest AIC and BIC values, indicating it provides the best balance between goodness-of-fit and model complexity.

### 2.3 Residual Diagnostics

After selecting the best model, we performed thorough residual diagnostics to ensure its validity.

#### 2.3.1 Coefficient Significance

Using the `coeftest()` function, we confirmed that both the AR and MA coefficients are statistically significant:

Coefficients:

|           | Estimate   | Std.Error | z value | Pr(> z )  |
|-----------|------------|-----------|---------|-----------|
| ar1       | 0.0662681  | 0.0994034 | 0.6667  | 0.5050    |
| ma1       | -0.5858463 | 0.0834042 | -7.0242 | 2.153e-12 |
| intercept | 0.0013768  | 0.0011764 | 1.1703  | 0.2419    |

Both coefficients are highly significant ( $p < 0.001$ ), supporting the appropriateness of the chosen model.

#### 2.3.2 Ljung-Box Test

We applied the Ljung-Box test at multiple lags to check for autocorrelation in residuals:

Ljung-Box Test (lag = 6 ): p-value = 0.2881146  
Ljung-Box Test (lag = 12 ): p-value = 0.005188962  
Ljung-Box Test (lag = 18 ): p-value = 0.0004847555

While the test does not reject independence at lag 6, we observe statistically significant autocorrelation at lags 12 and 18. This suggests that the current model may not fully capture the dependence structure in the data, and residual autocorrelation remains at higher lags. This could indicate that a more complex model or additional transformations might be necessary

## Appendix: R Script Used for Part I

Below is the complete R script (with comments) used to perform all operations